

DATA ANALYSIS INTERNSHIP

Day 1 – REPORT

The data analysis process contains five important steps:

1. Data Collection
2. Data Preprocessing
3. Exploratory Data Analysis (EDA)
4. Model Evaluation
5. Prediction

On Day 1, the concepts of Data Collection, Data Preprocessing, and Prediction were discussed and implemented using a Core Companies production dataset.

1. Data Collection

On Day 1, different dataset domains such as:

- Finance
- Transport
- Core Companies
- Software Companies

were discussed to create a dataset with 5 input columns and 1 output column.

From these topics, I selected **Core Companies** domain.

Created the following dataset structure:

Inputs

- Workers
- Machines
- Working Hours
- Raw Materials
- Electricity

Output

- Production Status

Added sample values for each input column.

Workers	Machines	Working Hours	Raw Materials	Electricity	Production Status
20 workers	5 machines	8 hours	100 kg steel	50 units	High
15 workers	4 machines	6 hours	40 kg iron	40 units	Low
10 workers	2 machines	4 hours	40 kg plastic	20 units	Low
25 workers	6 machines	9 hours	120 kg copper	60 units	High
12 workers	3 machines	5 hours	60 kg aluminium	30 units	Low

2. Data Preprocessing

Performed preprocessing to verify and organize the dataset.

Preprocessing Steps:

- Checked data types of each column
- Verified missing values
- Checked duplicate values
- Validated input formats

Data Types

Column	Data Type
Workers	Integer
Machines	Integer
Working Hours	Integer
Raw Materials	String
Electricity	String

Column	Data Type
Production Status	String

3. Prediction

Used the 5 input values to predict the output column.

Example 1

Inputs:

- Workers = 20
- Machines = 5
- Working Hours = 8
- Raw Materials = 100 kg steel
- Electricity = 50 units

Predicted Output:

- Production Status = High
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Example 2

Inputs:

- Workers = 10
- Machines = 2
- Working Hours = 4
- Raw Materials = 40 kg plastic
- Electricity = 20 units

Predicted Output:

- Production Status = Low

Conclusion:

On Day 1, I learned the basic concepts of data analysis such as data collection, preprocessing, and prediction. I created a core companies production dataset with 5 input columns and 1 output column. I also understood how preprocessing helps in organizing and validating data before analysis and prediction.

